

0.1 Section 6.4 - Binomial Identities

$$(a+b)^0 = 1$$

$$(a+b)^1 = 1a + 1b$$

$$(a+b)^2 = 1a^2 + 2a^1b^1 + 1b^2$$

$$(a+b)^3 = 1a^3 + 3a^2b^1 + 3a^1b^2 + 1b^3$$

$$(a+b)^4 = 1a^4 + 4a^3b^1 + 6a^2b^2 + 4a^1b^3 + 1b^4$$

$$(a+b)^5 = 1a^5 + 5a^4b^1 + 10a^3b^2 + 10a^2b^3 + 5a^1b^4 + 1b^5$$

$\vdots$

$$(a+b)^{10} = 1a^{10} + \underbrace{10}_{\binom{10}{9}} a^9b^1 + \underbrace{45}_{\binom{10}{8}} a^8b^2 + \underbrace{120}_{\binom{10}{7}} a^7b^3 + \underbrace{210}_{\binom{10}{6}} a^6b^4 + \underbrace{252}_{\binom{10}{5}} a^5b^5 + \underbrace{210}_{\binom{10}{4}} a^4b^6 + 120a^3b^7 + 45a^2b^8 +$$

For all of these, it's 10 choose either the power of a or the power of b . This is because the power is essentially the number of arrangements of $aaabbbbbbb$ for example

This brings us to our next theorem

0.1.1 Binomial Theorem

Theorem 1 (Binomial Theorem)

$$\text{For } n = 0, 1, 2, \dots, (a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

I would put the proof here but it's long and I'm lazy

Example 1 1. A pizza parlor has 10 toppings. How many pizzas are possible if any number of toppings can be chosen but none can be repeated?

$$\begin{aligned}
& \underbrace{\# \text{ 0 toppings}}_{\binom{10}{0}} + \underbrace{\# \text{ 1 topping}}_{\binom{10}{1}} + \dots + \underbrace{\# \text{ 0 toppings}}_{\binom{10}{10}} \\
&= \sum_{k=0}^{10} \binom{10}{k} \underbrace{1^{10-k} \cdot 1^k}_{\text{to use B.T}} \\
&= (1+1)^{10} \\
&= 2^{10} \\
&= \boxed{1024}
\end{aligned}$$

2. How many bit strings are there of length 10?

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^{10} = \boxed{1024}$$

or

Place k 0's in the bit string.

$$\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \dots + \binom{10}{10} = (1+1)^{10} = 2^{10} = 1024$$

3. How many subsets does a set with 10 elements have?

uhhhh same thing

0.1.2 Sum of Combinations

Theorem 2 (Sum of Combinations)

$$\text{For } n = 0, 1, 2, \dots, \sum_{k=0}^n \binom{n}{k} = 2^n$$

Example 2

uhhhhh just use the formula